

# CSCI461: Introduction to Big Data: Project Rubric

Spring2026

Note: Description of project phases is the baseline; examples are just a starting point to help you visualize the idea for your own project. You can adjust but you have to follow the basic outline of the project. Project Total Marks is 25.

## **1 Phase #1: Project Idea, Objective, and Dataset**

- Project Idea:

A well-constructed project idea is the foundation of your big data project. It should clearly articulate the problem you intend to address, the significance of this problem in the context of big data, and how your project aims to provide a solution or insight. Innovativeness and originality are key here, as it is a focus on real-world applicability. Provide a summary of the project's objectives and scope.

- Examples:

- \* Developing a real-time recommendation system for an e-commerce platform, utilizing Apache Hadoop for distributed data processing.
    - \* Analyzing healthcare data to predict disease outbreaks and improve resource allocation, employing Apache Spark for data analytics.
    - \* Creating a predictive maintenance system for manufacturing using sensor data and Apache Hadoop for data storage and processing.
    - \* Optimizing energy consumption in smart buildings through data-driven insights, leveraging Apache Spark for real-time analytics.
    - \* Implementing sentiment analysis on social media data to gauge public opinion, with Apache Hadoop managing the large dataset.

– Paper Section Reference:

Introduction: Introduce the project idea, its significance, and its alignment with broader big data challenges.

• Objective:

A good objective should be specific, measurable, achievable, relevant, and time-bound (SMART). Define what success looks like for your project and specify how you will measure it. Make sure the objectives align with the project idea and are realistically attainable within the given resources and timeline.

– Examples:

- \* To achieve a 20% increase in sales revenue within six months by delivering personalized product recommendations using Apache Spark.
- \* To reduce hospital readmission rates by 15% through early disease detection using Apache Hadoop for processing electronic health records.
- \* To decrease downtime in manufacturing by 30% by predicting machine failures using Apache Spark for predictive maintenance.
- \* To achieve a 15% reduction in energy costs by optimizing HVAC systems with real-time analytics powered by Apache Spark.
- \* To identify trends and influencers in public sentiment for marketing campaigns, leading to a 10% increase in engagement using Apache Hadoop for data analysis.

– Paper Section Reference:

Introduction: Include a clear problem statement and specific project objectives, along with success criteria.

• Dataset:

The choice of dataset is critical to the success of your project. Explain why you selected a particular dataset and how it relates to your project idea and objectives. Provide details on the dataset's source, size, format, and any preprocessing steps that may be required. Make sure the dataset is accessible and legally usable.

– Examples:

- \* Utilizing e-commerce transaction data from a company's database, stored in HDFS (Hadoop Distributed File System).

- \* Accessing electronic health records from a hospital's database, pre-processing and storing them in Apache Hadoop's HBase.
- \* Collecting sensor data from manufacturing equipment in real-time and storing it in HDFS.
- \* Gathering IoT data from smart buildings and storing it in Apache Kafka for subsequent analysis with Apache Spark.
- \* Extracting social media posts via web scraping and storing them in HDFS for sentiment analysis.

– Paper Section Reference:

Introduction: Mention the dataset's relevance, description, and source.

## **2 Phase #2: Project Planning, Design, Data Life-cycle, and Project Paper - Part 1**

- Project Planning:

In project planning, create a detailed roadmap for your big data project. This should include a project timeline with milestones, resource allocation (budget, hardware, software, personnel), and a risk assessment that identifies potential obstacles and mitigation strategies.

– Examples:

- \* Create a Gantt chart with clear timelines for data collection, preprocessing, modeling, and deployment, using Apache Hadoop and Apache Spark as central components.
- \* Allocate a budget for cloud-based Hadoop and Spark clusters, outlining the expected costs and resource scaling strategies.
- \* Identify risks such as data breaches, system failures, and budget overruns and outline mitigation plans, like encryption for data security.

– Paper Section Reference:

Methodology: Provide a comprehensive project plan that includes milestones, timelines, and resource allocation.

- Design:

Describe the architectural design of your big data solution. Explain why you chose specific technologies and how they align with your project objectives. Discuss scalability and flexibility considerations in your design.

– Examples:

- \* Design a distributed system with Apache Hadoop's MapReduce for data preprocessing and Apache Spark for real-time recommendation calculations.
- \* Select Apache Kafka for real-time data ingestion, ensuring compatibility with Spark for data processing.
- \* Ensure system scalability by using Hadoop's HDFS for data storage, accommodating large volumes of transaction data.

– Paper Section Reference:

Methodology: Elaborate on the system architecture and technology stack, justifying your choices.

• Data Lifecycle:

The data lifecycle is crucial in ensuring data quality and security. Explain how data is acquired, stored, processed, and visualized in your project. Include data flow diagrams and discuss measures taken to protect sensitive data.

– Examples:

- \* Acquire e-commerce transaction data through API integration, store it in HDFS, preprocess with Spark, and visualize trends with Apache Zeppelin.
- \* Collect electronic health records through secure channels, store them in Apache HBase, and use Spark for real-time analysis with Apache Spark Streaming.
- \* Implement data encryption and access controls in HDFS to secure sensitive manufacturing data, perform batch processing with MapReduce, and visualize insights with Apache Superset.

– Paper Section Reference:

Methodology: Detail the data acquisition, storage, processing, and visualization stages, emphasizing data security and privacy.

• Project Paper - Part 1:

In the IEEE-format paper's introduction, provide a clear and concise overview of your project. Include the problem statement, objectives, and a brief description of your project design and the dataset you'll be using.

– Examples:

- \* In the IEEE-format paper's introduction, provide an overview of the project's architecture, mentioning the use of Apache Hadoop for data storage and preprocessing and Apache Spark for real-time analytics.

– Paper Section Reference:

Introduction: Introduce the project idea, problem statement, objectives, and a high-level overview of the project's design.

### **3 Phase #3: Data Pre-Processing, Methodology, Contribution to the Community, and Project Paper - Part 2**

- Data Pre-Processing:

Data pre-processing is crucial for ensuring the quality and reliability of your data. Describe in detail the steps you took to clean and prepare your dataset. Address how you handle missing data, outliers, and noise, and explain any data transformation and feature engineering.

– Examples:

- \* Clean e-commerce transaction data by handling missing values and outliers, leveraging Apache Spark's DataFrame API.
- \* Impute missing health record values using Spark's MLlib, detect anomalies, and normalize data for analysis.
- \* Identify risks such as data breaches, system failures, and budget overruns and outline mitigation plans, like encryption for data security.
- \* Remove outliers from sensor data using Spark's statistical functions, transform data into appropriate formats for modeling.
- \* Aggregate IoT data into meaningful features, scaling values to a common range for clustering analysis.
- \* Preprocess social media text data by tokenization, removing stop words, and performing sentiment analysis using Spark's NLP libraries.

– Paper Section Reference:

Methodology: Provide a comprehensive description of data preprocessing steps, including methods for handling missing data, outlier detection, and feature engineering.

- **Methodology:**

In this section, go deep into the analytical methods and algorithms you employed. Explain why you chose specific methods, their implementation, and any optimizations you made. Include code snippets, parameters used, and any special considerations for big data.

- **Examples:**

- \* Utilize Apache Spark's MLlib for building a collaborative filtering recommendation model with ALS.
    - \* Employ Spark's GraphX for healthcare network analysis to detect disease spread patterns.
    - \* Implement a machine learning pipeline in Spark for predictive maintenance, combining feature engineering and ensemble models.
    - \* Apply Spark's MLlib to develop a time series forecasting model for energy consumption optimization.
    - \* Use Spark's NLP libraries for topic modeling and sentiment analysis of social media data.

- **Paper Section Reference:**

Methodology: Elaborate on the algorithms used, justify their selection, and provide implementation details.

- **Contribution to the Community:**

Explain how your project benefits the broader community or industry. If you make open-source contributions, discuss them. Address any ethical considerations and social impact. Provide evidence of how your work can be utilized or built upon by others.

- **Examples:**

- \* Contribute the recommendation model's source code and documentation to an open-source project, facilitating its adoption by other e-commerce platforms.
    - \* Organize workshops on healthcare analytics with Spark, sharing best practices and code snippets with the community.
    - \* Publish a white paper addressing ethical considerations in predictive maintenance and share insights on industry forums.
    - \* Collaborate with environmental organizations to provide energy optimization recommendations based on the project's findings.
    - \* Develop a sentiment analysis library for social media data processing in Spark and release it as open-source software.

- Paper Section Reference:

Discussion: Discuss the project's contribution to the community, including open-source contributions and ethical considerations.

- Project Paper - Part 2:

In the results and discussion section of the IEEE-format paper, present detailed findings from the Spark-based analysis, including visualizations, model performance metrics, and insights into the dataset. Discuss the broader implications of your work and its potential impact on the community and industry.

- Examples:

- \* In the results and discussion section of the IEEE-format paper, present your findings in a clear and structured manner. Interpret the results and discuss their significance in the context of your project's objectives and the broader field of big data.

- Paper Section Reference:

Results and Discussion: Present your findings and discuss their implications in detail.